

vision**2**learn

Understanding Coding Level 2

Hints
and
Tips

Unit 3



Contents

Section 1 – Know about different coding types	4
Section 2 – Understand what is meant by compiled and interpreted code	5
Section 3 - Understand what is meant by a pure function	6

Hints and Tips Unit 3

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 3: Understand coding design principles Level 2

Section 1 – Know about different coding types

Q1. In the table below, **define** what is meant by imperative and declarative coding.

This question covers both learning outcomes 1.1 and 1.2. For this you simply need to define what imperative and declarative coding mean. You may find it useful to give examples of each to demonstrate your understanding.

Q2. In the table below, **define** what is meant by the following coding types.

This question covers both learning outcomes 1.3 and 1.4. As with question 1, you simply need to define what functional and object-oriented coding are. What are the differences between the two? You may find it useful to give examples of each to demonstrate your understanding.

Q3. Do a little research to help you to **identify other coding types** in the table below. (Make sure you evidence websites and/or books used).

For this question you are being asked to identify other coding types. You may want to give examples of when you would use them.

Q4. In the table below, in your own words **state** the advantages and disadvantages of functional, object-orientated, and other coding types.

The first two types of coding are listed but to demonstrate understanding you should add as many different types as you can. For this question try to give as many advantages and disadvantages as you can for each of the different coding types.

Section 2 – Understand what is meant by compiled and interpreted code.

Q1. In the table below, **define the terms** compiled code and interpreted code.

This question is simply asking for a definition of the two different types of code. What does each mean? You should give examples to demonstrate your understanding.

Q2. In the table below, **explain the principles** of compiled code and interpreted code.

For this question you need to think about the principles of the two different types of code. How do operate? This is an explain question so try to provide good detail in your answer.

Q3. In the table below, **Explain** the advantages and disadvantages of compiled code and interpreted code.

For this question try to give as many advantages and disadvantages as you can for each of the different coding types.

Section 3 - Understand what is meant by a pure function

Q1. In the table below, **define the terms** pure function and impure function.

For this question you simply need to define each term. What does it mean? You may find it useful to give examples to demonstrate understanding.

Q2. In the table below, **identify** potential dependencies and/or needs in pure function..

This is an identify question but you need to make it clear what each potential dependency or need is. try to be as detailed as possible in your answer.

Q3. In the box below, **explain** the benefits of pure function.

This is an explain question so try and provide good detail in your response. Think about why you would use pure function in coding. What benefits does it have over impure functions?

Q4. In the box below, **explain** the term parallelisation.

This is an explain question so try and provide good detail in your response. What is meant by parallelisation, what does it do?

Q5. In the box below, **describe** how parallelisation is managed

This is a describe question so try and provide good detail in your response.

Think about how systems manage parallelisation. How does it work. Try to give an example to demonstrate your understanding.